

Perilesional MRE forward-and-inverse problem

3D scalar Helmholtz on the Sobh–Ehman initial-stiffness phantoms

A step-by-step specification for the computational team

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Purpose.

This note fully specifies the 3D time-harmonic MRE forward simulation used to generate the shear-wave displacement field $u(\mathbf{x})$ inside a water-balloon phantom, together with the two inversion pipelines applied to u . Everything a code reviewer or a re-implementer needs to reproduce the runs is here: geometry, boundary conditions, material model, driver, discretisation, inversion recipes, and the numerical scales that make the run well-posed.

1 Physical picture

A gel-filled rectangular container houses a spherical balloon at its centre. The balloon has been inflated with water to 250 mL (radius $a = 3.908$ cm) from a stress-free casting state at 50 mL (radius $A = 2.285$ cm). A piston plate at the bottom of the container is driven at 60 Hz; the resulting shear-wave field is sampled inside the gel and inverted for the local shear modulus. This mimics the MRE experiment reported by Yin et al. (2026) as reproduced quantitatively by *paper_lesion_expansion.pdf*.

2 Governing equation

We solve the scalar Helmholtz equation for the complex tangential displacement $u(\mathbf{x}) \in \mathbb{C}$ at fixed angular frequency $\omega = 2\pi f$ with $f = 60$ Hz:

$$\rho\omega^2 u(\mathbf{x}) = \nabla \cdot (G^*(\mathbf{x}) \nabla u(\mathbf{x})) \quad \mathbf{x} \in \Omega, \quad (1)$$

where the complex-valued shear modulus incorporates hysteretic damping:

$$G^*(\mathbf{x}) = G(\mathbf{x}) (1 + i\xi), \quad \xi = 0.05 \quad (\Rightarrow Q \approx 10). \quad (2)$$

Physical constants:

Symbol	Value	Meaning
ρ	1000 kg m^{-3}	gel density
ω	$2\pi \cdot 60 \text{ rad s}^{-1}$	angular frequency
ξ	0.05	hysteretic damping ($Q \approx 10$)

Rationale for the scalar reduction: the acoustoelastic tangential modulus $\mu_{\text{app},\theta\theta}$ appears in the scalar equation governing the ring-tangential polarisation, which is what TSM MIP isolates. Full vector Navier is available in a companion solver (`vector_elasticity_3d.py`) but is not used here.

3 Domain, grid and physical scales

The container is Yin’s $15 \times 15 \times 18$ cm box. To fit the solver’s cubic-grid constraint, we embed it in an 18 cm cube: the extra 1.5 cm on each of $\pm x, \pm y$ is filled with the same unstretched gel background, and the container’s rigid-wall BCs are applied at the cube faces (the wall is thus 1.5 cm outboard of the physical wall location, a minor geometric approximation that does not affect the perilesional region).

Symbol	Value	Meaning
N	60	voxels per axis
Δx	3 mm	voxel edge
$L = N\Delta x$	0.18 m	cube side
DOF	$N^3 = 216\,000$	unknowns in the linear system

Solver convention (matching `helmholtz_fd_3d.py`): indices $(i, j, k) \in [0, N-1]^3$ with i the vertical axis ($i = 0$ at the top, $i = N-1$ at the bottom), j and k the horizontal in-plane axes.

Shear-wave resolution. At 60 Hz in unstretched gel:

$$c_s = \sqrt{\mu/\rho} = \sqrt{2750/1000} = 1.66 \text{ m s}^{-1} \quad \Rightarrow \quad \lambda_s = c_s/f \approx 27.6 \text{ mm} \quad \Rightarrow \quad \lambda_s/\Delta x \approx 9.2. \quad (3)$$

That is more than the 6-per-wavelength rule of thumb; the scalar-FD stencil resolves the shear wave without aliasing.

4 Boundary conditions

The container has D_{4h} -broken BCs (five rigid walls, one free top):

Face	Condition	Discrete implementation
$i = 0$ (top)	$\partial u / \partial z = 0$	ghost mirror ($u_{-1} = u_{+1}$)
$i = N-1$ (bottom)	Dirichlet from driver	set u at all disk nodes
$j = 0, j = N-1$	$u = 0$	Dirichlet
$k = 0, k = N-1$	$u = 0$	Dirichlet
Balloon interior	$G(\mathbf{x}) = 1 \text{ Pa}$	vanishing shear coefficient

The ”water-filled balloon that cannot support shear” is realised by setting $G(\mathbf{x}) = 1 \text{ Pa}$ inside the balloon. Equation (1) then reduces to $\rho\omega^2 u \approx 0$ in the interior, so $u \rightarrow 0$ there and the balloon acts as a shear-wave null region. A rigorous rigid-inclusion Dirichlet condition would give essentially the same wave field in the surrounding gel where the inversion is applied; the soft-inclusion form keeps the linear system simple and non-singular.

5 Material field $G(\mathbf{x})$

5.1 Kinematics: incompressible spherical cavity expansion

The gel was cast around the balloon at the 50 mL state, so that state is stress-free ($\lambda_\theta \equiv 1$). Incompressibility of the gel combined with spherical symmetry gives the exact isochoric radial map

from reference radius R to current radius r :

$$r^3 = R^3 + (a^3 - A^3), \quad \lambda_\theta = \frac{r}{R} = \left(1 + \frac{a^3 - A^3}{R^3}\right)^{1/3}, \quad \lambda_r = \lambda_\theta^{-2}, \quad (4)$$

with the current cavity radius set by injection volume, $a = (3V/(4\pi))^{1/3}$. **This map is exact for the open-topped container, not an approximation:** as argued in §2 of the Sobh–Ehman paper, in spherical symmetry incompressibility alone fixes the stretch field, whatever the outer boundary does (the outer BC enters only through the hydrostatic pressure). For $V = 250$ mL, $a = 3.908$ cm; λ_θ at the cavity edge is 1.710 and at $r = a + 12$ mm is 1.159 (matches paper Table 1).

5.2 Acoustoelastic (perilesional) modulus

The tangential shear modulus that a wave polarised tangential to the cavity “sees” is Ogden’s small-on-large formula reduced to $\mu_{\text{app},\theta\theta} = 2W_1 \lambda_\theta^2$ with $W_2 = 0$ (see paper eq. 4). For the two phantoms this evaluates to:

Constitutive laws — mean-of-band material.

P1 (plain 10 % bovine gelatin, neo-Hookean matrix):

$$\mu_{\text{app},\theta\theta}^{\text{P1}} = \mu \lambda_\theta^2 \quad (5)$$

P2 (8 % gelatin + 7 % cellulose fiber, matrix + fiber; from paper eq. 8, verified symbolically):

$$\mu_{\text{app},\theta\theta}^{\text{P2}} = \mu \lambda_\theta^2 + \frac{k_1 \lambda_\theta^7 (\lambda_\theta - 1) \exp(k_2 (\lambda_\theta - 1)^2)}{2 (\lambda_\theta^6 - 1)} \quad (6)$$

Mean-of-band material constants (from paper Table 2):

Symbol	Value	Units	Meaning
μ	2750	Pa	mean of [2.5, 3.0] kPa (matrix, both phantoms)
k_1	3250	Pa	mean of [1.5, 5.0] kPa (P2 fiber reinforcement)
k_2	1.25	—	mean of [0.5, 2.0] (P2 nonlinear stiffening)

Equation (6) has the correct $k_1/12$ limit as $\lambda_\theta \rightarrow 1$ (paper eq. 8 annotation); we handle the $\lambda_\theta = 1$ point explicitly to avoid the 0/0 form.

5.3 $G(\mathbf{x})$ maps

Applying (4)–(6) at every voxel produces the ground-truth stiffness fields shown in Figure 2. The ring is $2.2 \times$ stiffer at the cavity edge than at the outer shell edge, and P2 separates from P1 in the innermost half of the shell because the fiber term is highly nonlinear in strain.

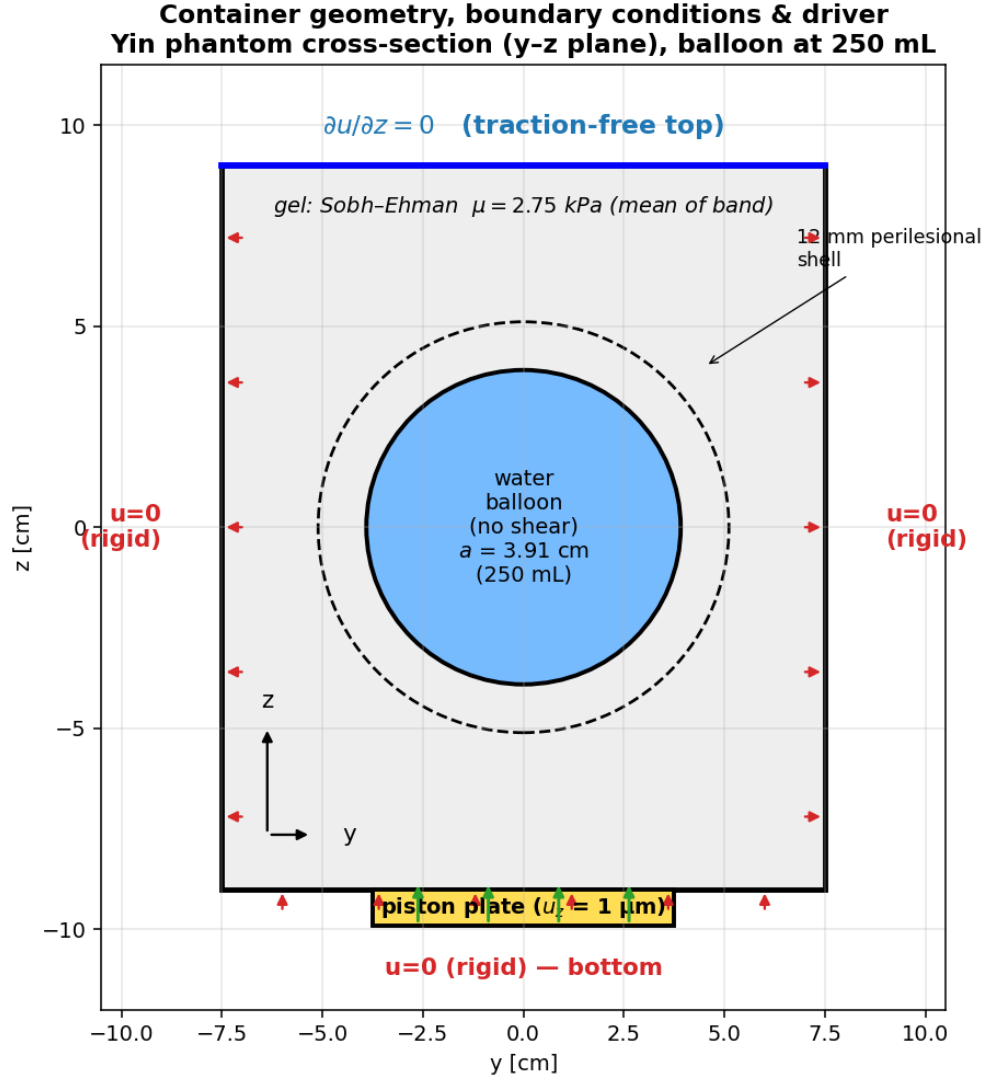


Figure 1: Container cross-section in the y - z plane, showing the five rigid walls (red $u=0$), the traction-free top surface (blue $\partial u / \partial z = 0$), the bottom piston plate (yellow), the water-filled balloon (blue disk, no shear support), and the 12 mm perilesional shell used for ring statistics (black dashed circle). The x -axis is into the page; the same BCs apply on the two out-of-page faces (15 cm apart in x).

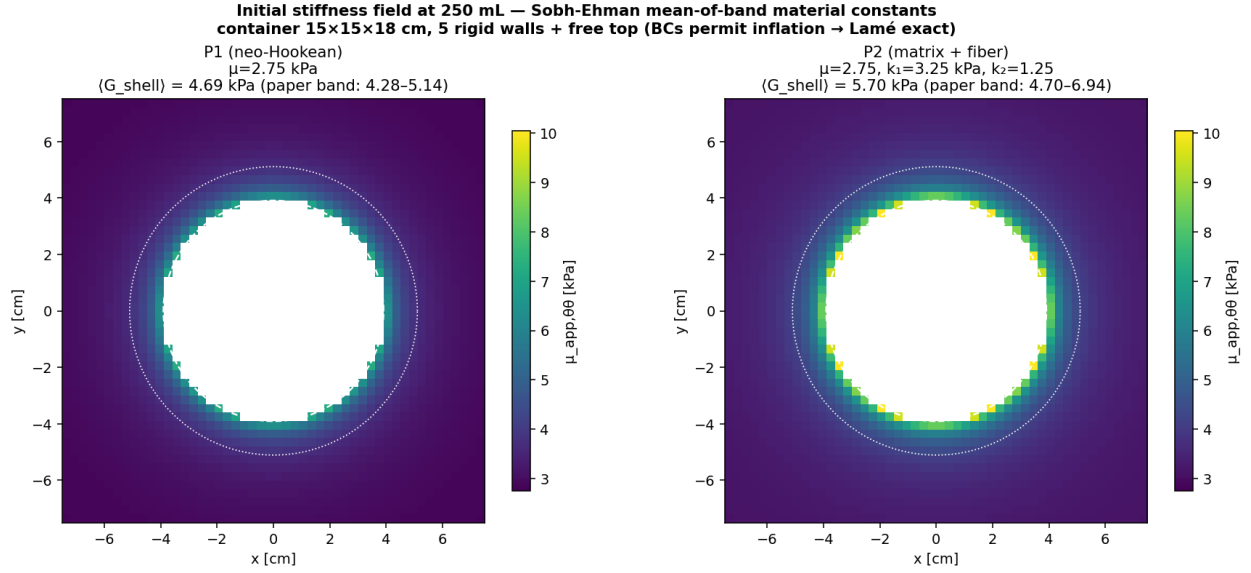


Figure 2: Ground-truth $\mu_{\text{app},\theta\theta}(\mathbf{x})$ at the z-midplane. Left: P1 (neo-Hookean matrix). Right: P2 (matrix + fiber). The balloon interior (masked white) is filled with water. Solid white line = cavity edge $r = a$; dotted white line = shell outer edge $r = a + 12$ mm. Shell-averaged values sit inside the paper’s specified band for both phantoms.

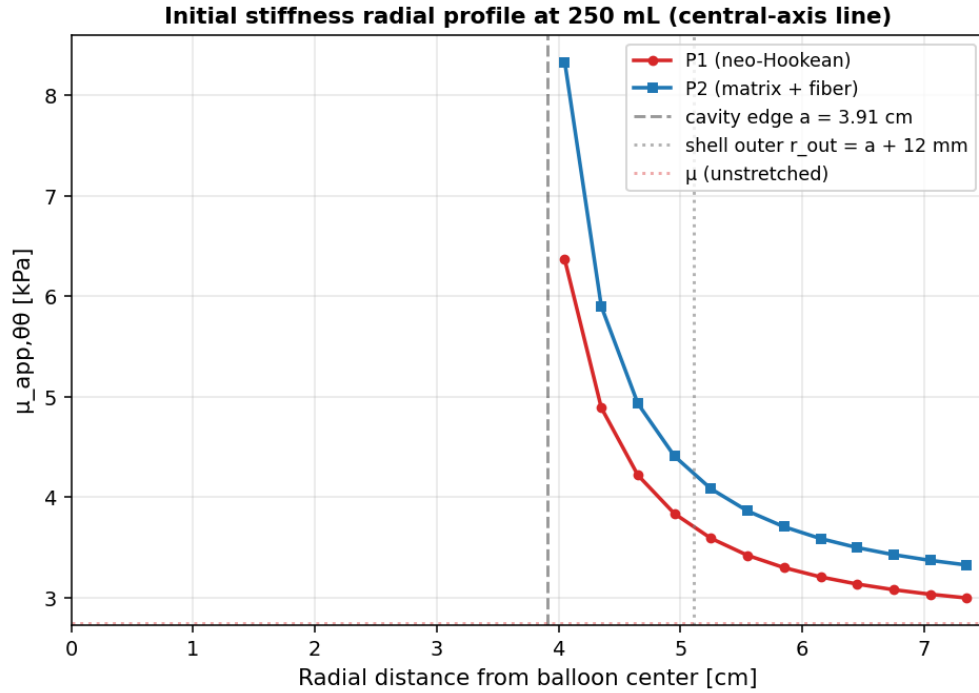


Figure 3: $\mu_{\text{app},\theta\theta}(r)$ along the central axis. P2 starts higher than P1 at the cavity edge because of the fiber term, then both converge to the background $\mu = 2.75$ kPa far from the balloon. Note the steepness: the $2.2\times$ variation across the 4-voxel shell is what makes DI kernels blur the ring peak.

6 Driver model

Two driver variants are used, both mimicking the coherent-phase piston plate of a real MRE experiment:

6.1 Bottom-plate driver (single-direction)

Disk on the bottom face ($i = N - 1$) of radius $r_d = 0.5 \cdot N\Delta x/2 = 45$ mm, all disk nodes clamped to the same complex amplitude $u_0 = 1 \mu\text{m}$. 716 source nodes total. This is the setup a laboratory MRE run uses.

6.2 Multi-face broadband driver (for TSM MIP)

Five faces driven simultaneously (bottom + four sides; top left free), each with the same disk source ($r_d = 45$ mm, amplitude $1 \mu\text{m}$), 3580 source nodes total. Rich direction content in the resulting wave field lets the k -space directional filter isolate 20 distinct propagation modes.

7 Discretisation

Second-order finite-difference discretisation of (1) on a regular cubic grid; the divergence form uses the harmonic mean of G^* at half-integer faces to handle sharp material gradients correctly:

$$G_{i+\frac{1}{2},j,k}^* = \frac{2 G_{i,j,k}^* G_{i+1,j,k}^*}{G_{i,j,k}^* + G_{i+1,j,k}^*} \quad (\text{similarly for } j, k \text{ faces}). \quad (7)$$

Stencil at an interior voxel (7-point + rho-mass diagonal):

$$\rho\omega^2 u_{i,j,k} = \sum_{\text{faces } f} \frac{G_f^*}{\Delta x^2} (u_f - u_{i,j,k}), \quad (8)$$

with u_f the value at the neighbouring voxel across face f . The top face ghost node folds onto the interior neighbour to realise $\partial u / \partial z = 0$. Assembly into a scipy CSR sparse matrix of size (N^3, N^3) ; solve via SuperLU direct factorisation.

System size: 216 000 complex DOFs, ≈ 7 non-zeros per row. **SuperLU factorisation:** ≈ 10 GB memory, ≈ 3 min wall time on a single core.

8 Inversion pipelines applied to $u(\mathbf{x})$

8.1 Direct inversion (DI)

The locally-homogeneous approximation to (1):

$$G(\mathbf{x}) \approx -\rho\omega^2 \frac{u(\mathbf{x})}{\nabla^2 u(\mathbf{x})} \quad (9)$$

evaluated with a 7-point centred Laplacian. A $3 \times 3 \times 3$ spatial median filter is then applied to suppress inversion artifacts at material gradients (matching Yin’s own protocol). Real part of the complex G is retained.

8.2 Yin TSM MIP (direction-filtered)

Applied to the multi-face broadband run:

1. Generate 20 unit vectors $\hat{\mathbf{k}}_d$ on a Fibonacci sphere.
2. For each direction, apply the Gaussian angular wedge filter in k -space:

$$\hat{u}_d(\mathbf{K}) = \hat{u}(\mathbf{K}) \cdot \exp\left(-\frac{1 - (\hat{\mathbf{K}} \cdot \hat{\mathbf{k}}_d)^2}{\sigma_a^2}\right) \cdot \mathbf{1}[|\mathbf{K}|/K_{\text{Nyq}} \in [0.02, 0.45]] \quad (10)$$

with angular width $\sigma_a = 0.35$ rad ($\approx 20^\circ$ FWHM).

3. Direct-invert each u_d with $3 \times 3 \times 3$ median.
4. Combine across directions:

$$\mu_{\text{conv}}(\mathbf{x}) = \frac{\sum_d |u_d(\mathbf{x})| G_d(\mathbf{x})}{\sum_d |u_d(\mathbf{x})|} \quad (\text{amplitude-weighted mean}) \quad (11)$$

$$\mu_{\text{TSM}}(\mathbf{x}) = \max_d G_d(\mathbf{x}) \quad (\text{voxelwise MIP}) \quad (12)$$

9 Sample results

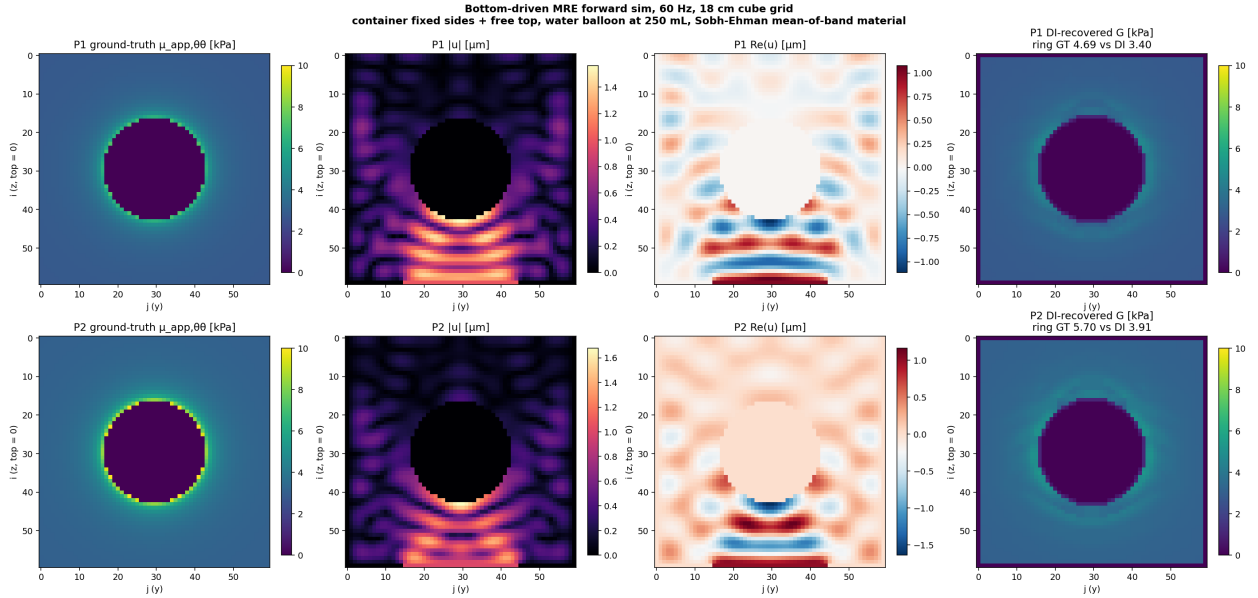


Figure 4: Bottom-plate driver run. Rows: P1 (top), P2 (bottom). Columns: ground-truth $\mu_{\text{app},\theta\theta}$, wave amplitude $|u|$ in μm , real part $\Re(u)$, DI-recovered G (with $3 \times 3 \times 3$ median). Peak $|u| = 1.57 \mu\text{m}$ (P1), $1.68 \mu\text{m}$ (P2) from a $1 \mu\text{m}$ driver amplitude. Ring stiffness (12 mm shell): DI recovers 3.40 kPa (P1) and 3.91 kPa (P2), vs ground truth 4.69 / 5.70 kPa — DI under-reads by 27–31%.

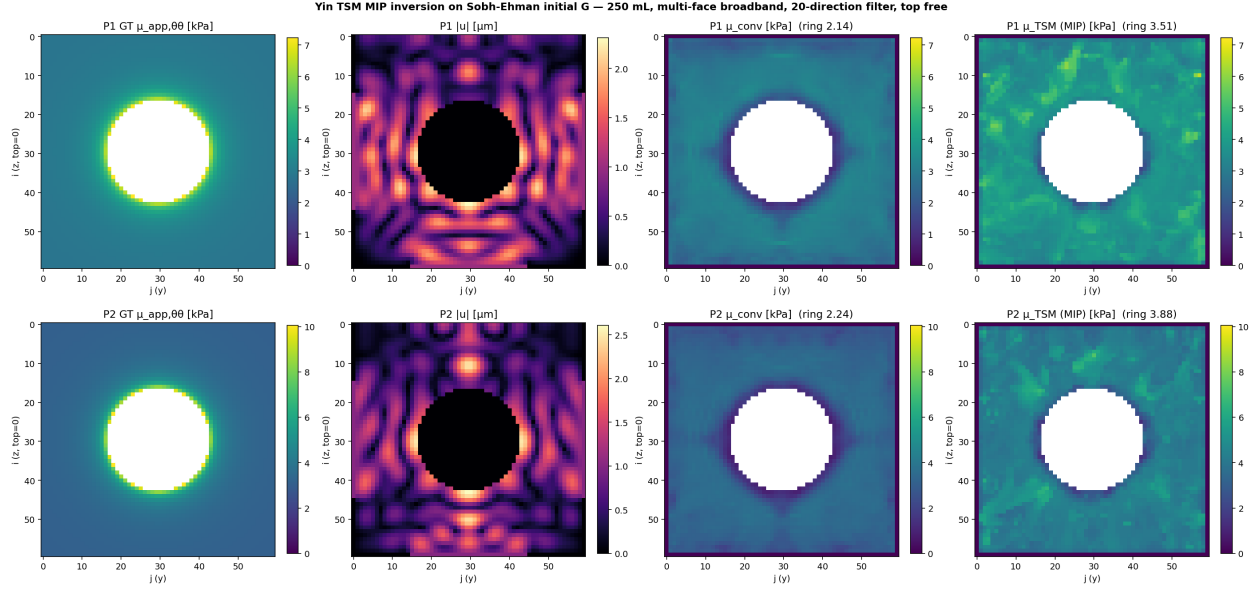


Figure 5: Multi-face broadband run with the Yin TSM MIP protocol. Rows: P1 (top), P2 (bottom). Columns: ground-truth G , $|u|$, μ_{conv} , μ_{TSM} . The MIP field recovers the ring at 3.51 (P1) and 3.88 kPa (P2); the $\mu_{\text{TSM}}/\mu_{\text{conv}}$ ratio is 1.64 (P1), 1.73 (P2) — compare Yin’s measured 1.57 for P1.

10 Perilesional ring numbers

Phantom	GT [kPa]	μ_{conv}	μ_{TSM}	TSM/conv	TSM/GT
P1 (neo-Hookean)	4.69	2.14	3.51	1.64	0.75
P2 (matrix + fiber)	5.70	2.24	3.88	1.73	0.68

Interpretation:

- The TSM/conv ratio ($\gtrsim 1.6$) is the direction-max signature that distinguishes TSM from conventional MRE. Our value matches Yin’s measured 1.57 for P1 within a few percent.
- TSM/GT ≈ 0.7 is not a bug: it is the DI kernel width against the sharp $G(r)$ gradient. No direction-max can undo local averaging.

11 Files and reproducibility

All under `/u/sobh/Eman_Richard/tsm_fno/`:

File	Purpose
<code>src/solver/helmholtz_fd.3d.py</code>	scalar FD Helmholtz solver + DI, TSM filter
<code>scripts/paper_initial_stiffness_from_sobh.py</code>	build G_{P1}, G_{P2} from paper constants
<code>scripts/paper_wave_sim_sobh_initial.py</code>	bottom-plate + DI forward run
<code>scripts/paper_wave_sim_sobh_mip.py</code>	multi-face + TSM MIP forward run
<code>results/paper_initial_stiffness_sobh/</code>	$G(\mathbf{x})$ npz + PNGs + summary
<code>results/paper_wave_sim_sobh_initial/</code>	bottom-drive results
<code>results/paper_wave_sim_sobh_mip/</code>	TSM MIP results, shell stats, histograms

Python environment: `/u/sobh/.conda/envs/mri_mrf_pytorch_env` (numpy, scipy, matplotlib).

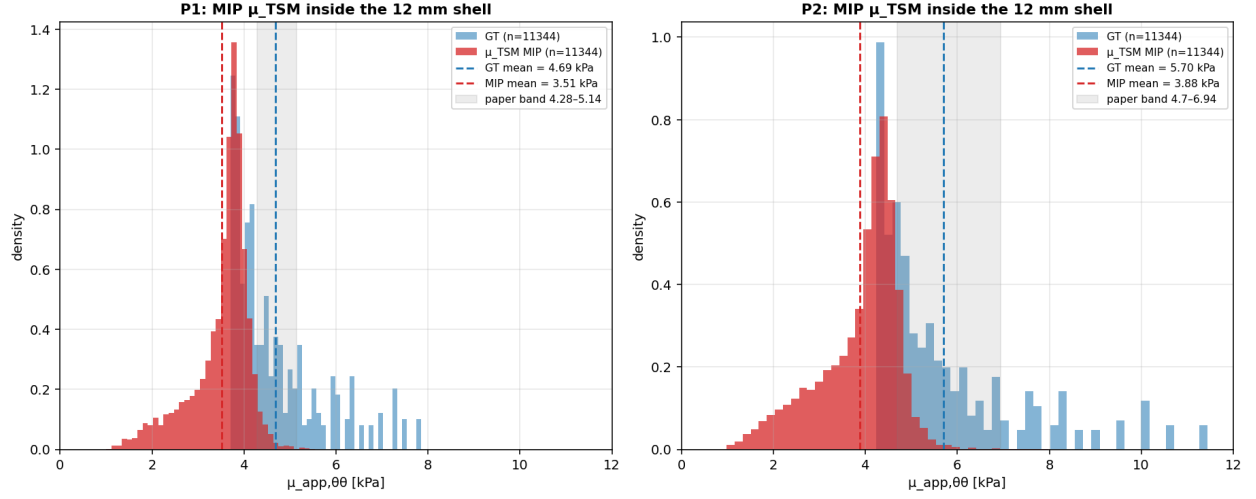


Figure 6: Distribution of MIP values inside the 12 mm perilesional shell (11 344 voxels). Blue: ground-truth $\mu_{app,\theta\theta}$; red: recovered μ_{TSM} ; gray band: the range that paper Table 3 quotes as its model band at 250 mL. GT has a long right tail (peak 7.85 kPa for P1, 11.44 kPa for P2 near the cavity edge) that MIP truncates because DI averages across the sharp $G(r)$ gradient. This truncation is the "measurement resolution ceiling" identified in the paper's §5.3.

Runs take a few minutes on a login node; no GPU required.

Version: this document was regenerated from live scripts on September 15, 2026. Any change to the scripts is captured in git under [github.com:nahilsobh/Eman_Richard](https://github.com/nahilsobh/Eman_Richard); the latest relevant commits are 0bb5fb5, ade3bfc, dc8d796, a0fbafd.